SWIGGY SALES ANALYSIS

Import Libraries

In [1]:
```
import pandas as pd  # to cleaning, aggration in data
import numpy as np    # to do complex mathematical operations
import matplotlib.pyplot as plt # to create different charts (basic level charts)
import seaborn as sns    # advance version of matplotlib
import plotly.express as px   # filtering charts
```

In [2]:
```
!pip install plotly
```

```
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: plotly in .\AppData\Roaming\Python\Python313\site-packages (6.6.0)
Requirement already satisfied: narwhals>=1.15.1 in .\AppData\Roaming\Python\Python313\site-packages (from plotly) (2.16.0)
Requirement already satisfied: packaging in .\AppData\Roaming\Python\Python313\site-packages (from plotly) (24.2)
```

Import Data

In [3]:
```
df = pd.read_excel(r"C:\Users\3568s\Downloads\swiggy_data.xlsx")
```

In [4]:
```
df.head(10)
```

Out[4]:

	State	City	Order Date	Restaurant Name	Location	Category	Dish Name	Price (INR)	Rating	Rating Count
0	Karnataka	Bengaluru	2025-06-29	Anand Sweets & Savouries	Rajarajeshwari Nagar	Snack	Butter Murukku-200gm	133.9	4.0	0
1	Karnataka	Bengaluru	2025-04-03	Srinidhi Sagar Deluxe	Kengeri	Recommended	Badam Milk	52.0	4.5	25
2	Karnataka	Bengaluru	2025-01-15	Srinidhi Sagar Deluxe	Kengeri	Recommended	Chow Chow Bath	117.0	4.7	48
3	Karnataka	Bengaluru	2025-04-17	Srinidhi Sagar Deluxe	Kengeri	Recommended	Kesari Bath	65.0	4.6	65
4	Karnataka	Bengaluru	2025-03-13	Srinidhi Sagar Deluxe	Kengeri	Recommended	Mix Raitha	130.0	4.0	0
5	Karnataka	Bengaluru	2025-07-08	Srinidhi Sagar Deluxe	Kengeri	Recommended	Srinidhi Sagar Special	312.0	4.0	0
6	Karnataka	Bengaluru	2025-01-21	Srinidhi Sagar Deluxe	Kengeri	Recommended	Garlic Naan	98.0	4.0	34
7	Karnataka	Bengaluru	2025-04-13	Srinidhi Sagar Deluxe	Kengeri	Recommended	Pista	137.0	4.0	0
8	Karnataka	Bengaluru	2025-05-02	Srinidhi Sagar Deluxe	Kengeri	North Indian Gravy	Panneer Butter Masala	241.0	4.4	29
9	Karnataka	Bengaluru	2025-07-30	Srinidhi Sagar Deluxe	Kengeri	North Indian Gravy	Dal Tadka	195.0	4.9	51

In [5]:
```
df.tail(10)
```

Out[5]:

	State	City	Order Date	Restaurant Name	Location	Category	Dish Name	Price (INR)	Rating	Rating Count
197420	Sikkim	Gangtok	2025-02-09	Mama's Kitchen	Gangtok	Momos	Mushroom cheese steam momo ...	112.0	4.4	0
197421	Sikkim	Gangtok	2025-07-28	Mama's Kitchen	Gangtok	Momos	Mushroom cheese fried momo ...	126.0	4.4	0
197422	Sikkim	Gangtok	2025-03-17	Mama's Kitchen	Gangtok	Momos	Mushroom cheese chilli momo ...	140.0	4.4	0
197423	Sikkim	Gangtok	2025-06-18	Mama's Kitchen	Gangtok	Momos	Soya cheese steam momo ...	84.0	4.4	0
197424	Sikkim	Gangtok	2025-08-11	Mama's Kitchen	Gangtok	Momos	Soya cheese fried momo ...	98.0	4.4	0
197425	Sikkim	Gangtok	2025-01-25	Mama's Kitchen	Gangtok	Momos	Soya cheese chilli momo ...	112.0	4.4	0
197426	Sikkim	Gangtok	2025-07-02	Mama's Kitchen	Gangtok	Momos	Kurkure momo fried ...	140.0	4.4	0
197427	Sikkim	Gangtok	2025-03-25	Mama's Kitchen	Gangtok	Momos	Chilli cheese momo	126.0	4.4	0
197428	Sikkim	Gangtok	2025-03-26	Mama's Kitchen	Gangtok	Momos	Veg Momos (8 Pc)	85.0	4.4	0
197429	Sikkim	Gangtok	2025-03-27	Mama's Kitchen	Gangtok	Momos	Soya Momo	100.0	4.4	0

MetaData

In [6]:
```
print("Number of Rows:", df.shape[0])   # Number of rows
```

```
Number of Rows: 197430
```

In [7]:
```
print("No of Fields:", df.shape[1])   # Number of columns
```

```
No of Fields: 10
```

In [8]:
```
df.info     # sampel data, identifying data
```

Out[8]:

```
<bound method DataFrame.info of            State       City Order Date           Restaurant Name  \
0       Karnataka  Bengaluru 2025-06-29  Anand Sweets & Savouries
1       Karnataka  Bengaluru 2025-04-03     Srinidhi Sagar Deluxe
2       Karnataka  Bengaluru 2025-01-15     Srinidhi Sagar Deluxe
3       Karnataka  Bengaluru 2025-04-17     Srinidhi Sagar Deluxe
4       Karnataka  Bengaluru 2025-03-13     Srinidhi Sagar Deluxe
...           ...        ...        ...                       ...
197425     Sikkim    Gangtok 2025-01-25            Mama's Kitchen
197426     Sikkim    Gangtok 2025-07-02            Mama's Kitchen
197427     Sikkim    Gangtok 2025-03-25            Mama's Kitchen
197428     Sikkim    Gangtok 2025-03-26            Mama's Kitchen
197429     Sikkim    Gangtok 2025-03-27            Mama's Kitchen

                    Location     Category  \
0       Rajarajeshwari Nagar        Snack
1                   Kengeri  Recommended
2                   Kengeri  Recommended
3                   Kengeri  Recommended
4                   Kengeri  Recommended
...                     ...          ...
197425              Gangtok        Momos
197426              Gangtok        Momos
197427              Gangtok        Momos
197428              Gangtok        Momos
197429              Gangtok        Momos

                                            Dish Name  Price (INR)  \
0                                   Butter Murukku-200gm        133.9
1                                             Badam Milk         52.0
2                                        Chow Chow Bath        117.0
3                                           Kesari Bath         65.0
4                                            Mix Raitha        130.0
...                                                 ...          ...
197425  Soya cheese chilli momo                     ...        112.0
197426  Kurkure momo fried                          ...        140.0
197427                               Chilli cheese momo        126.0
197428                                  Veg Momos (8 Pc)         85.0
197429                                         Soya Momo        100.0

        Rating  Rating Count
0          4.0             0
1          4.5            25
2          4.7            48
3          4.6            65
4          4.0             0
...        ...           ...
197425     4.4             0
197426     4.4             0
197427     4.4             0
197428     4.4             0
197429     4.4             0
```

```
[197430 rows x 10 columns]>
```

Data Types

In [9]:
```
df.dtypes     # checking the all datatypes
```

Out[9]:
```
State                     object
City                      object
Order Date         datetime64[ns]
Restaurant Name            object
Location                   object
Category                   object
Dish Name                  object
Price (INR)               float64
Rating                    float64
Rating Count                int64
dtype: object
```

In [10]:
```
df.describe()   # all information
```

Out[10]:

	Order Date	Price (INR)	Rating	Rating Count
count	197430	197430.000000	197430.000000	197430.000000
mean	2025-05-01 19:41:20.996808960	268.512920	4.341582	28.321805
min	2025-01-01 00:00:00	0.950000	1.500000	0.000000
25%	2025-03-01 00:00:00	139.000000	4.300000	0.000000
50%	2025-05-02 00:00:00	229.000000	4.400000	2.000000
75%	2025-07-01 00:00:00	329.000000	4.500000	15.000000
max	2025-08-31 00:00:00	8000.000000	5.000000	999.000000
std	NaN	219.338363	0.422585	87.542593

KPI's

Total Sales

In [11]:
```
total_sales = df["Price (INR)"].sum()
print("Total Sales (INR):", round(total_sales,2))
```

```
Total Sales (INR): 53012505.77
```

Average Rating

In [12]:
```
average_rating = df["Rating"].mean()
print("Average Rating:", round(average_rating,1))
```

```
Average Rating: 4.3
```

Average Order Value

In [13]:
```
average_order_value = df["Price (INR)"].mean()
print("Average Order Value (INR):", round(average_order_value,2))
```

```
Average Order Value (INR): 268.51
```

Ratings Count

In [14]:
```
ratings_count = df["Rating Count"].sum()
print("Rating Count:", round(ratings_count,2))
```

```
Rating Count: 5591574
```

Total Orders

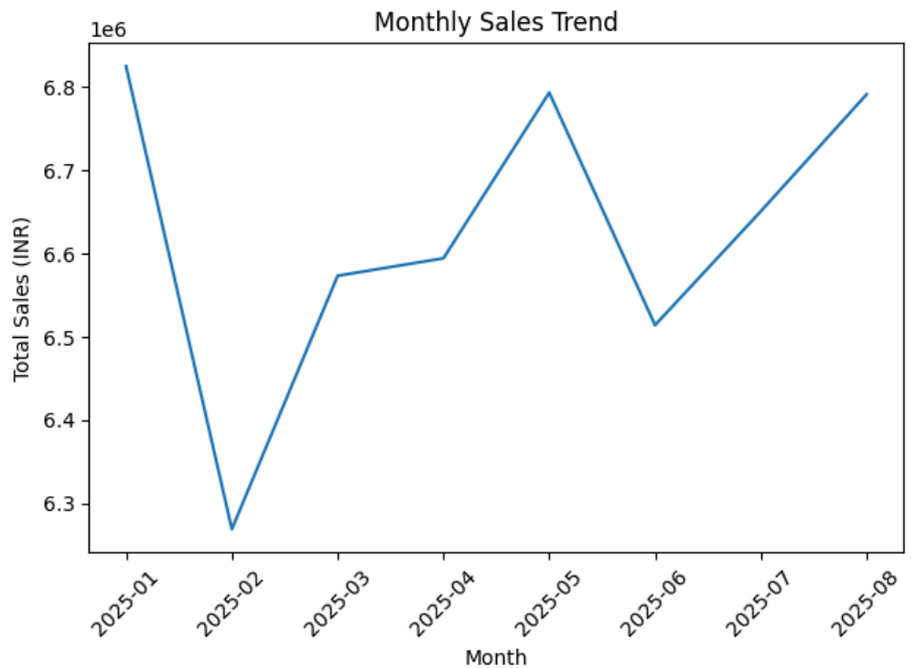
In [15]:
```
total_orders = len(df)
print(total_orders)
```

```
197430
```

Charts Design

Monthly Sales Trend

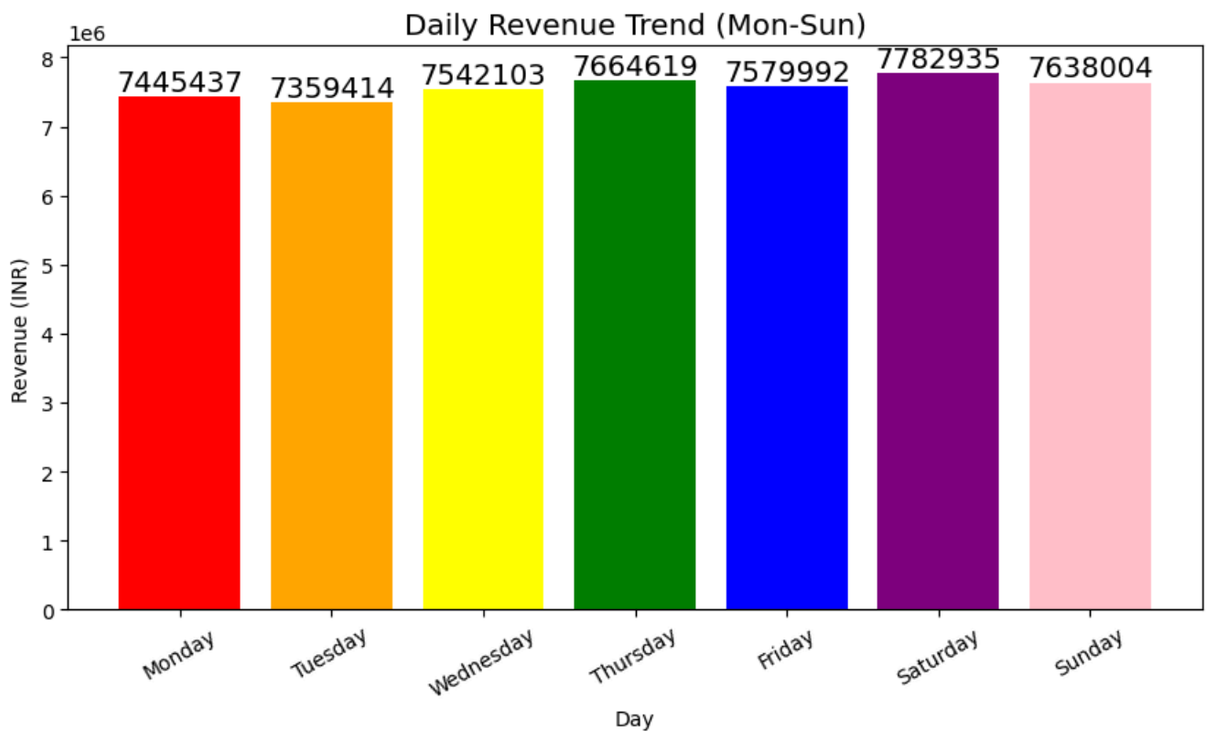
In [16]:
```
# df["Order Date"] = pd.to_datetime = (df["Order Date"])

df["YearMonth"] = df["Order Date"].dt.to_period("M").astype(str)

monthly_revenue = df.groupby("YearMonth")["Price (INR)"].sum().reset_index()

plt.figure()
plt.plot(monthly_revenue["YearMonth"], monthly_revenue["Price (INR)"])
plt.xticks(rotation=45)
plt.xlabel("Month")
plt.ylabel("Total Sales (INR)")
plt.title("Monthly Sales Trend")
plt.tight_layout()
plt.show()
```



Daily Sales Trend

In [17]:
```
df.columns = df.columns.str.strip()

df["Order Date"] = pd.to_datetime(df["Order Date"])

df["DayName"] = df["Order Date"].dt.day_name()

daily_revenue = (
    df.groupby("DayName")["Price (INR)"]
    .sum()
    .reindex(["Monday","Tuesday","Wednesday","Thursday","Friday","Saturday","Sunday"])
)

plt.figure(figsize=(10,5))
plt.bar(daily_revenue.index, daily_revenue.values)
plt.style.use("ggplot")
plt.title("Daily Revenue Trend (Mon-Sun)")
plt.xlabel("Day")
plt.ylabel("Revenue (INR)")
plt.xticks(rotation=30)
```

```
plt.style.use("fivethirtyeight")

bars = plt.bar(daily_revenue.index, daily_revenue.values, color=["red","orange","yellow","green","blue","purple","pink"])

# plt.bar(daily_revenue.index, daily_revenue.values, color=colors)


for bar in bars:
    yval = bar.get_height()
    plt.text(bar.get_x() + bar.get_width()/2, yval, int(yval),
             ha='center', va='bottom')

plt.show()
```



Total Sales by Food Type (Veg vs Non-Veg)

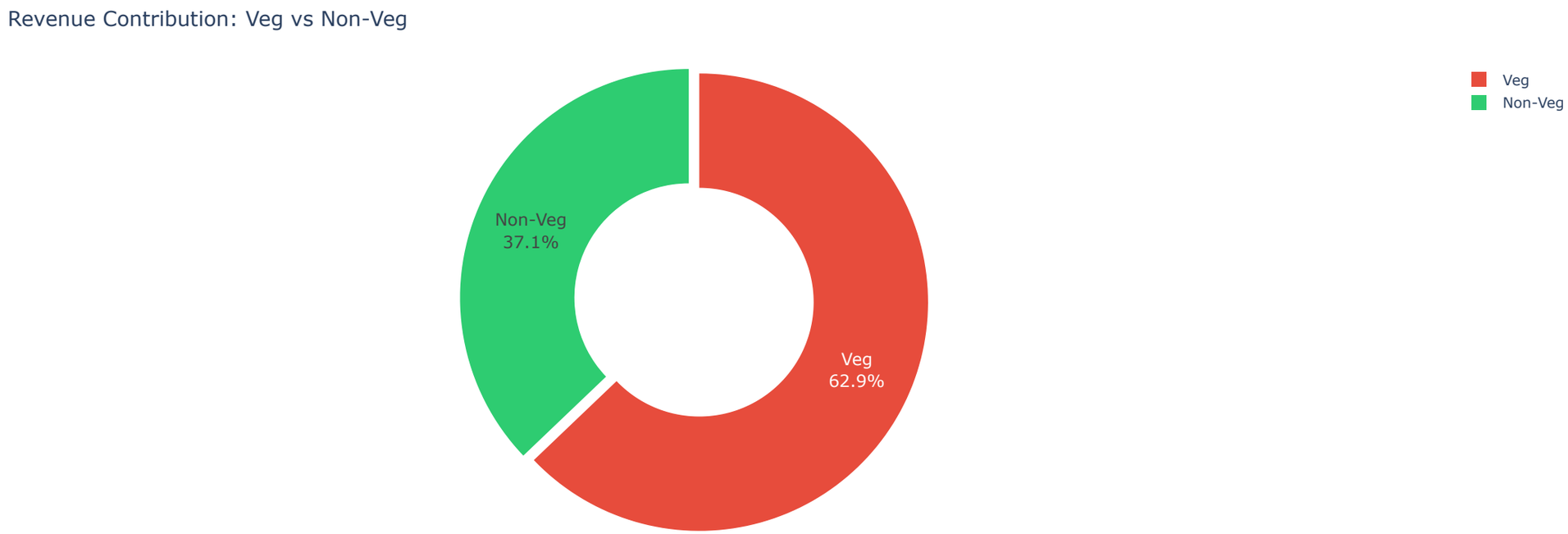
In [18]:
```
non_veg_keywords = [
    "chicken","egg","fish","mutton",
    "prawn","biryani","kabab","kebab",
    "non-veg","non veg"
```

```
]

df["Food Category"] = np.where(
    df["Dish Name"].str.lower().str.contains("|".join(non_veg_keywords), na = False),
    "Non-Veg",
    "Veg"
)
```

In [19]:
```
food_revenue = (
    df.groupby("Food Category")["Price (INR)"]
    .sum()
    .reset_index()
)
```

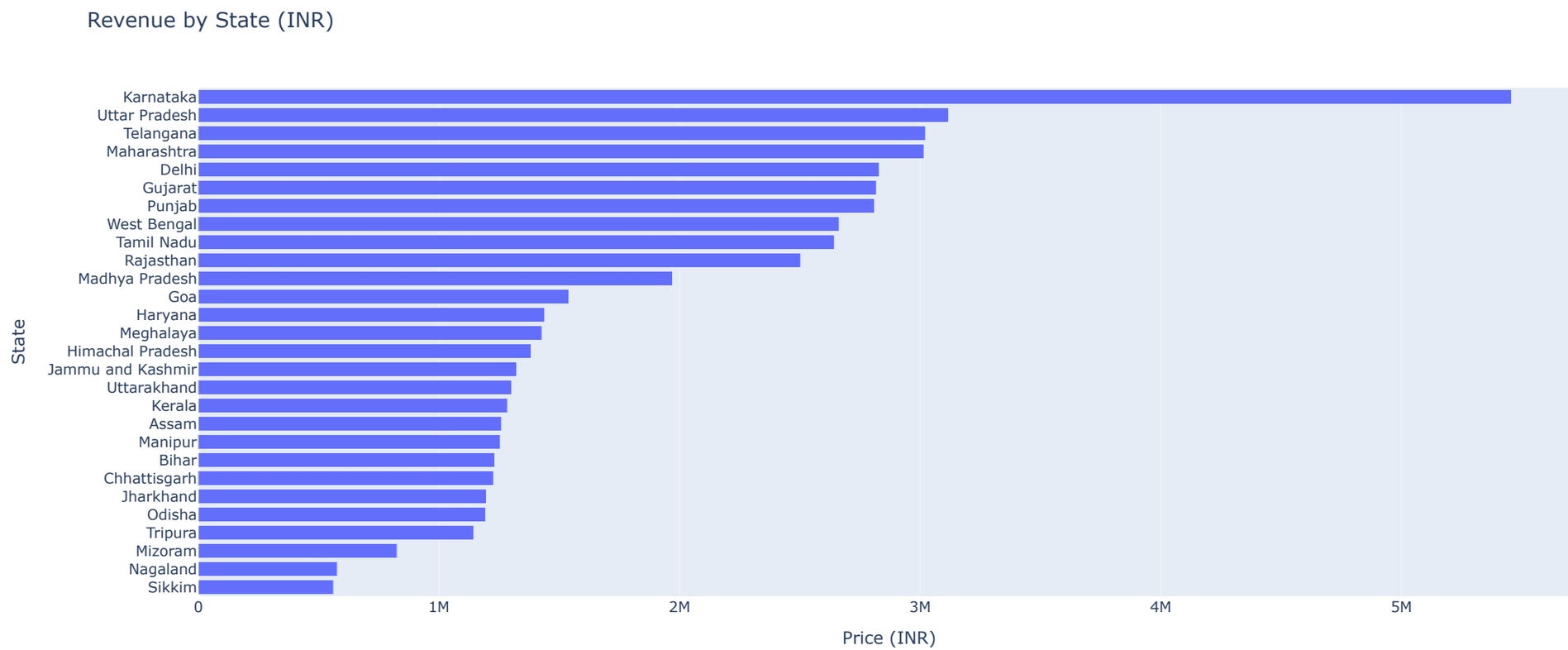
In [20]:
```
import plotly.io as pio
pio.renderers.default = "notebook"
```

In [21]:
```
import plotly.express as px

food_revenue = df.groupby("Food Category")["Price (INR)"].sum().reset_index()

fig = px.pie(
    food_revenue,
    values="Price (INR)",
    names="Food Category",
    hole=0.5,
    title="Revenue Contribution: Veg vs Non-Veg",
)

fig.update_traces(
    textinfo="percent+label",
    pull=[0.05, 0]
)

fig.update_layout(
    height=500,
    margin=dict(t=60,b=40,l=40,r=40)
)


fig.update_traces(
    textinfo="percent+label",
    textfont_size=14,
    marker=dict(colors=["#2ECC71","#E74C3C"]),
    pull=[0.05,0]
)

fig.show()
```



Total Sales by State

In [22]:
```
fig = px.bar(
    df.groupby("State", as_index=False)["Price (INR)"].sum()
    .sort_values("Price (INR)", ascending=False),
    x = "Price (INR)",
    y = "State",
    orientation = "h",
    title = "Revenue by State (INR)"
)

fig.update_layout(height=600, yaxis=dict(autorange="reversed"))
fig.show()
```



Quarterly Performance Summary

In [23]:
```
df["Order_Date"] = pd.to_datetime(df["Order Date"])
df["Quarter"] = df["Order_Date"].dt.to_period("Q").astype(str)

quarterly_summary = (
    df.groupby("Quarter", as_index=False)
    .agg(
        Total_Sales=("Price (INR)", "sum"),
        Avg_Rating=("Rating", "mean"),
        Total_Orders=("Order_Date", "count")
    )
    .sort_values("Quarter")
)
```

```
quarterly_summary["Total_Sales"] = quarterly_summary["Total_Sales"].round(0)
quarterly_summary["Avg_Rating"] = quarterly_summary["Avg_Rating"].round(2)

quarterly_summary
```

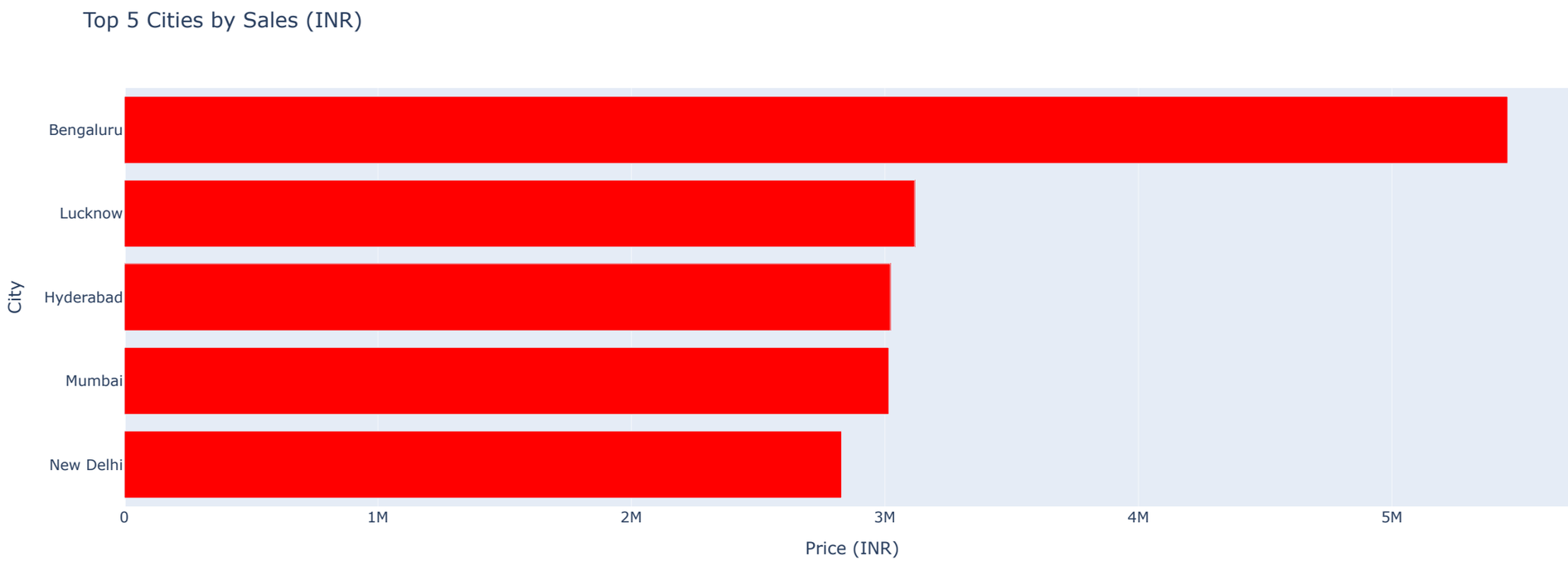
Out[23]:

	Quarter	Total_Sales	Avg_Rating	Total_Orders
0	2025Q1	19667822.0	4.34	73096
1	2025Q2	19902257.0	4.34	74163
2	2025Q3	13442427.0	4.34	50171

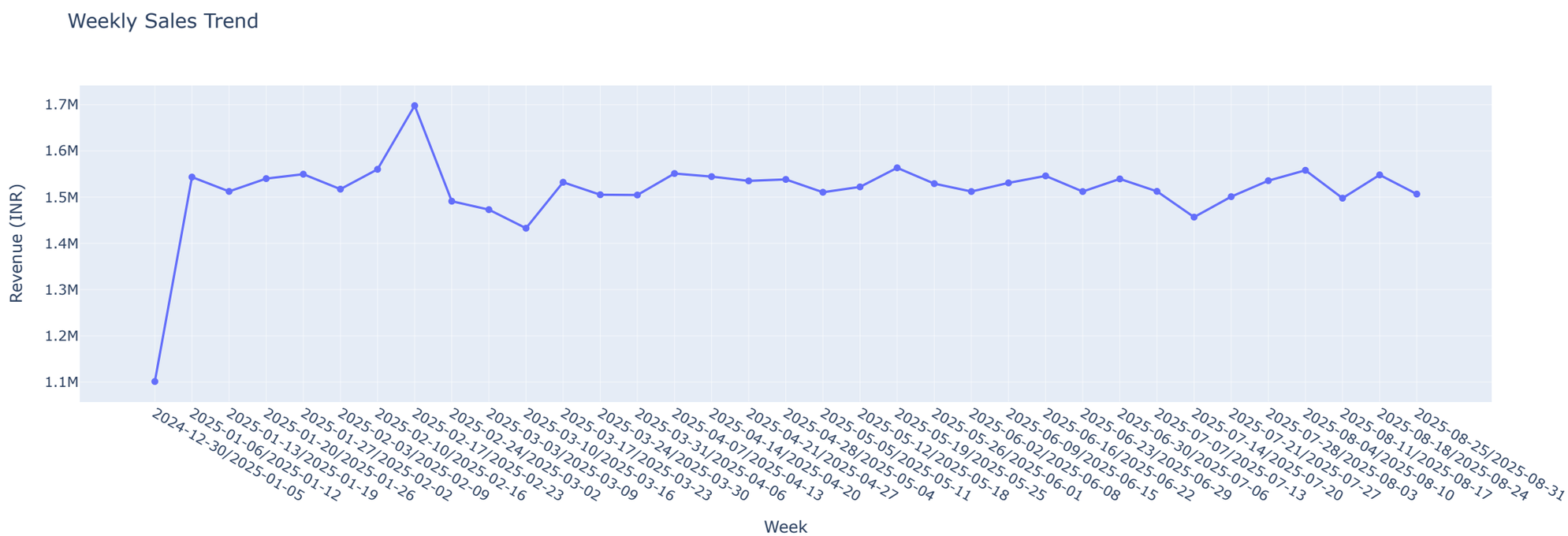
Top 5 Cities by Sales

In [24]:

```
top_5_cities = (
    df.groupby("City")["Price (INR)"]
    .sum()
    .nlargest(5)
    .sort_values()
    .reset_index()
)

fig = px.bar(
    top_5_cities,
    x="Price (INR)",
    y="City",
    orientation="h",
    title="Top 5 Cities by Sales (INR)",
    color_discrete_sequence=["red"]
)

fig.show()
```



Weekly Trend Analysis

In [25]:
```
df["Order_Date"] = pd.to_datetime(df["Order Date"])

df["Week"] = df["Order_Date"].dt.to_period("W").astype(str)

weekly_sales = (
    df.groupby("Week", as_index=False)["Price (INR)"]
    .sum()
)

fig = px.line(
    weekly_sales,
    x="Week",
    y="Price (INR)",
    title="Weekly Sales Trend",
    markers=True
)
```

```python
fig.update_layout(
    xaxis_title="Week",
    yaxis_title="Revenue (INR)",
    height=500
)

fig.show()
```